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Inclusion-Exclusion

– General

For $n=2$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For $n=3$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Generally for n sets

If $A_1, A_2, \dots, A_n$ are n finite sets, then:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{1 \leq i \leq n} |A_i| \\ &\quad - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ &\quad - \dots \\ &\quad + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

– Inclusion-Exclusion of properties

For x elements we have n properties $P_1, P_2, \dots, P_n$. We say that P'_i is the negation of property P_i .

The number of elements satisfying all properties P_i, P_j, P_k is $N(P_i, P_j, P_k)$

The number of elements that satisfy *none* of the properties is:

$$\begin{aligned} N(P'_1, P'_2, \dots, P'_n) &= N - \sum_{1 \leq i \leq n} N(P_i) \\ &\quad + \sum_{1 \leq i < j \leq n} N(P_i, P_j) \\ &\quad - \sum_{1 \leq i < j < k \leq n} N(P_i, P_j, P_k) \\ &\quad + \dots \\ &\quad + (-1)^n N(P_1, P_2, \dots, P_n) \end{aligned}$$

where N is the total number of elements.

— Derangements

A derangement is a permutation that leaves no object in its original position. The amount of derangements of n objects, denoted by $!n$, is given by:

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right]$$